Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	The University of Auckland
Short name / acronym used in projects	UoA
Participant Identification Code (PIC), if available	999613131
Website(s)	https://profiles.auckland.ac.nz/i-rupenthal; www.botu.nz (note, group website is currently undergoing update and may not be available)
Country	New Zealand
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Private Bag 92019, Auckland 1142, New Zealand
Department(s), institute(s), research group(s) or unit(s) that would be involved	Buchanan Ocular Therapeutics Unit, Department of Ophthalmology
Additional institutional resources relevant to collaboration	Cell culture suite, confocal microscope, animal facility, access to human tissues via the New Zealand National Eye Bank

Main contact person for this profile	Ilva Rupenthal, Professor, Director of the Buchanan Ocular Therapeutics Unit, Department of Ophthalmology, Faculty of Medical and Health Sciences
Main contact email	i.rupenthal@auckland.a.cnz

Other contact person(s), if relevant	Dr Priyanka Agarwal, pharmaceutical formulation lead, p.agarwal@auckland.ac.nz Dr Lola Mugisho, biomedical lead, lola.mugisho@auckland.ac.nz

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	x Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	x Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): in vitro and ex vivo models

Free Keywords for internal searchable database (please provide up to 10)	Drug delivery; drug penetration; formulation; inflammation; inflammasome; extracellular vesicles; human tissue explants; in vitro and ex vivo models
Geographies of interest for exchanges (optional)	NA

Overview of your research / innovation focus
The BOTUT team focusses on translational research integrating pharmaceutical and biomedical expertise to develop innovative ocular drug delivery systems with improved safety and efficacy with current projects on dry eye, diabetic retinopathy and uveitis treatment. We have worked on nanoparticulate, polymeric, in situ gelling and semifluorinated alkane systems for the delivery of various therapeutics but specifically inflammasome blockers to reduce inflammatory processes in various ocular diseases. We also have extensive expertise in preclinical models used to translate novel ocular therapeutics from bench to bedside, including ocular drug penetration models as well as in vitro disease models using cells and tissue explants.

Why your team could be relevant to a translational eye-health and vision consortium
Our team would bring specific expertise in developing and testing ocular delivery systems, evaluating the efficacy of novel treatments in our inflammatory in vitro and ex vivo models, providing expertise in inflammasome biology and most importantly offering clinically relevant human explant models that support clinical translation of novel therapies. We use our ocular models as a window to neurodegenerative diseases and have started to investigate the involvement of the gut-eye-axis in chronic inflammatory eye diseases. Finally, we have extensive expertise in research commercialization being inventors and income sharing scientist on patents as well as being involved in two university spinout companies.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
• *Ex vivo ocular drug penetration models and analytical capabilities (fluorescence/confocal microscope and HPLC)* • *In vitro cell models of diabetic retinopathy and other inflammatory retinal conditions*

- *Human organotypic retinal culture (sandwich model including retina, choroid and sclera) to investigate disease pathways and novel drug molecules*
- *Human choroidal explant model to investigate neovascularisation processes and novel therapies*
- *Animal models of ocular and neurodegenerative diseases (although limited)*

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Cell culture suite including a CELLCYTE X live cell imager and analyzer*
- *Fluorescence and confocal microscope*
- *Navicyte Ussing chamber with electrodes*
- *Access to human tissues from the New Zealand National Eye Bank*
- *In depth grading system for experimental autoimmune uveitis signs (and collaboration on AI algorithms for automated grading)*
- *Animal facility with limited ocular and neurodegenerative disease models*
- *Access to clinical cohorts via our clinicians*

Training capacity and knowledge-transfer potential

- *Technical/scientific/translational: In vitro and ex vivo models for drug penetration and anti-inflammatory drug efficacy*
- *Entrepreneurial: patenting and founding a start-up company*

What your team would hope to gain from participating

Sharing our models to be used by others; access to other models to test our inflammasome therapeutics; joint publications and longer-term strategic collaboration; insight into the European research culture and funding system; industrial insight and regulatory know-how

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Ilva Rupenthal	Professor and Director of the Buchanan Ocular Therapeutics Unit	Ophthalmology	Researcher	Experienced	i.rupenthal@auckland.ac.nz
Lola Mugisho	Senior Research Fellow	Ophthalmology	Researcher	Experienced	lola.mugisho@auckland.ac.nz
Priyanka Agarwal	Senior Research Fellow	Ophthalmology	Researcher	Experiences	p.agarwal@auckland.ac.nz

Main disciplines represented in your team	Pharmaceutical sciences, biomaterials, cell biology, immunology and inflammation, commercialization
Experience in EU / international / multi-partner collaborative projects	Alexander von Humboldt fellowship (Cologne, Germany), DAAD PPP grant (Marburg, Germany), ERASMUS+ (Tarsus, Turkey) and FCMH (Marburg/Giessen. Germany) fellowships; multi-partner Health Research Council of New Zealand project and program grant involvement
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note: Translational/preclinical research; patient facing clinical work only via clinical collaborators

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
1. *SHOME A, MUGISHO OO, AGARWAL P, NIEDERER RL,* ***RUPENTHAL ID*** *(2026) Blocking the inflammasome pathway: An effective way to treat inflammation in experimental autoimmune uveitis, European Journal of Pharmacology 1011:178442 DOI: 10.1016/j.ejphar.2025.178442*
2. *NASCIMENTO BARBALHO G, BRUGGER S, GRATIERI T, KECK CM, RUPENTHAL ID, AGARWAL P (2026) Influence of simulated tear flow on ocular penetration of transfersomes, European Journal of Pharmaceutics and Biopharmaceutics, DOI: 10.1016/j.ejpb.2026.115154*
3. *LOUIE HH, MUGISHO OO, CHAMLEY LW, RUPENTHAL ID (2025) Small extracellular vesicles released from ARPE-19 cells grown under diabetic retinopathy conditions promote NLRP3 inflammasome activation, ACS Biomaterials Science & Engineering 11(20):6134-6149 DOI: 10.1021/acsbiomaterials.5c00718*
4. *MUGISHO OO, ARYAL J, SHOME A, LYON H, ACOSTA ML, GREEN CR, RUPENTHAL ID (2023) Orally delivered connexin43 hemichannel blocker, tonabersat, inhibits vascular breakdown and inflammasome activation in a mouse model of diabetic retinopathy, International Journal of Molecular Sciences 24(4):38764 DOI: 10.3390/ijms24043876*
5. *KUO YJ, LOUIE HH, RUPENTHAL ID, MUGISHO OO (2021) Characterization of a novel human organotypic retinal culture technique, Journal of Visualized Experiments 172:e620464 DOI: 10.3791/62046*

Up to 5 relevant previous projects, grants, networks or collaborations
1. *Solving chronic and recurrent inflammation: Beyond corticosteroid therapy, Health Research Council Explorer Grant, (NZ$150,000, principal investigator, 2026-2028)* national
2. *Blocking the inflammasome pathway in neurodegenerative diseases, Buchanan Charitable Foundation (NZ$1,800,000, co-principal investigator, 2025-2031)* national with international collaborations in the future
3. *Gut-eye axis – is the inflammasome the missing link? Safe Sight Society Research Grant (NZ$17,135, principal investigator, 2024-2025)* national with international collaboration
4. *Humboldt Research Fellowship for Experienced Researchers, Tackling the vicious circle of dry eye disease (€37,210, principal investigator, 2022-2024)* international with the University of Cologne
5. *Health Research Council Project Grant, Tackling the vicious circle of dry eye disease (NZ$1,195,447, principal investigator, 3 years)* national with international collaboration

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
1. *SHOME A, MUGISHO OO, RUPENTHAL ID. Improved treatment of inflammation. International Patent Application No. PCT/IB2025/059738 (filed 25th September 2025)*
2. *COUTINHO FP, RUPENTHAL ID, GREEN CR. Methods of treatment and novel constructs. International Patent Application No. PCT/NZ2018/050059 (submitted 28th April 2018); now at National Phase US11466069B2, CN111201247A, EP3615575A4, WO2018199777A1, CA3061738A1, JP2024059834A*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host

Administrative / HR / legal considerations that may affect mobility	visa

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
• *Expertise in developing and characterizing ocular formulations, especially for poorly soluble drugs* • *Extensive in vitro and ex vivo models to test anti-inflammatory drugs with strong inflammasome biology expertise* • *Access to human donor tissues to investigate certain disease pathways*

Capabilities or expertise you would especially like to access through the consortium
• *Access to other relevant animal models to test anti-inflammasome drugs* • *Gut-eye-axis research capabilities (e.g. microbiome or gut tissue preparation expertise)* • *Immunology expertise (e.g. flow cytometry analysis of draining lymph nodes)* • *Long-term international collaborations*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ Academic ☐ Clinical ☐ SME/industry ☐ NGO/foundation ☐ Public body ☐ Other

Organisation / team name:

Country:

Website:

Existing link with your team: ☐ Current collaborator ☐ Previous collaborator ☐ Desired collaborator ☐ Other

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Universidad de Valladolid

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Ilva Rupenthal
Date completed: 22/08/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)